

ES 120 COMSOL Project: Testing Two Assumptions of the Mechanics of Solids

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1 Introduction

This project tests two simplifying assumptions used throughout ES 120 by running finite element simulations in COMSOL on a standard ASTM E8 round tensile specimen:

- (I) **Linearization of elasticity.** ES 120 uses a linear theory that drops nonlinear terms from the elasticity equations on the assumption that strains are small.
- (II) **Saint-Venant’s principle and idealized stress distributions.** ES 120 uses simple stress formulas (like $\sigma = P/A$) that assume uniform stress on every cross-section. Real boundary loads are never perfectly uniform, but Saint-Venant’s principle says these formulas still work far enough away from where the load is applied.

2 Approach

2.1 Common Setup

The geometry is a standard ASTM E8 round tensile specimen (total length $L = 145$ mm, gauge length 50 mm, gauge diameter 12.5 mm, grip diameter $D_C = 20$ mm, fillet radius 10 mm). The material is A36 structural steel ($E = 200$ GPa, $\nu = 0.30$). The lower end face is fixed, loads are applied to the upper end face along the bar’s axis, and all simulations are stationary (steady-state) on a tetrahedral mesh.

I first checked the setup against an ES 120 hand calculation at $\sigma_{\text{app}} = 200$ MPa ($F_{\text{app}} = 62.83$ kN). The hand calculation gives a gauge stress of 512 MPa and a tip displacement of 0.232 mm. COMSOL gave 512 MPa in the gauge (Figure 1) and a tip displacement of 0.250 mm. The small discrepancy in displacement comes from fillet compliance that the simple three-cylinder formula doesn’t capture.

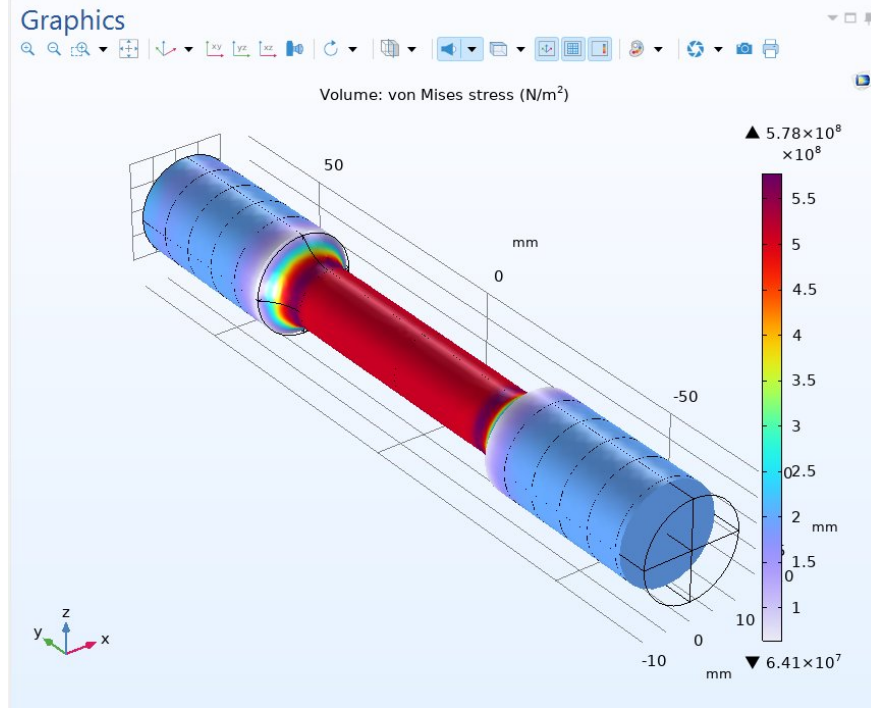


Figure 1: Von Mises stress at the reference load. The gauge section (red middle) is at 512 MPa, the grip ends are at 200 MPa, and the peak 578 MPa occurs at the fillets.

I then ran a mesh convergence check at five mesh sizes from *Coarser* (~ 1500 elements) to *Finer* (~ 41000 elements). Tip displacement varied by only 0.084% across that range (Figure 2), so the results are mesh-converged. I used the *Fine* setting (13923 elements, ~ 5 s solution time) for all following simulations.

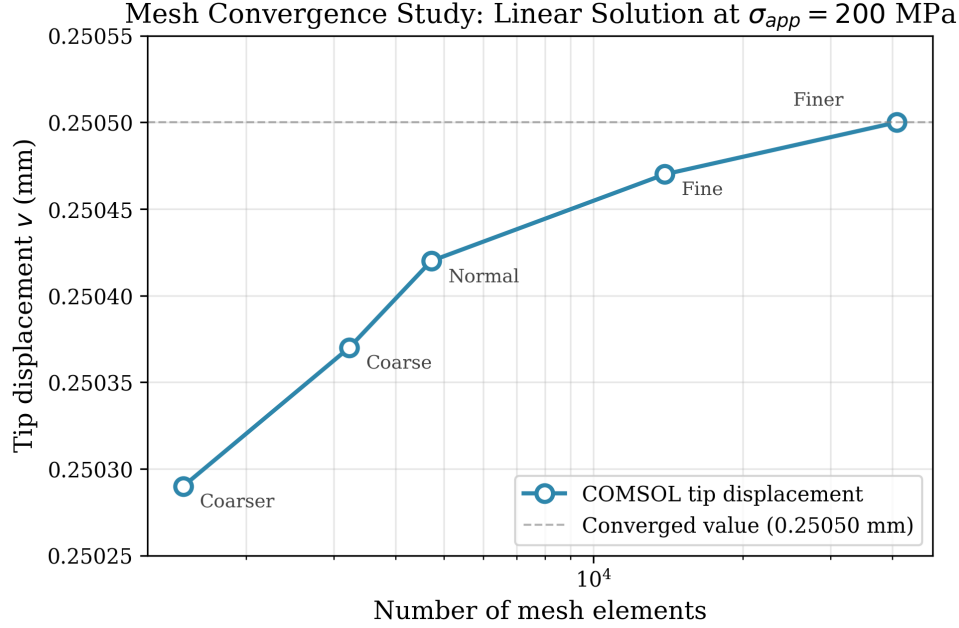


Figure 2: Mesh convergence: tip displacement varies by less than 0.1% across five mesh refinements.

2.2 Assumption I: Linear vs. Nonlinear Sweep

COMSOL has a single checkbox for “Include geometric nonlinearity” that toggles between the linearized small-deformation theory used in ES 120 and the full nonlinear theory of elasticity. I ran a parametric sweep over σ_{app} at eight log-spaced values from 20 MPa to 60 GPa (corresponding to nominal strains $\varepsilon = \sigma/E$ from 10^{-4} to 0.3), once with nonlinearity on and once with it off. The metric is the tip displacement v of the loaded end face, and the quantity I compared is the percent disagreement between the two solutions.

The high strain end of this sweep is unrealistic for real steel (yield strain is around 1.25×10^{-3}), but since the material model is kept linear elastic throughout, this is a mathematical probe of the linearization assumption itself, not a model of real material behavior.

2.3 Assumption II: Three Statically Equivalent Loadings

I applied three different boundary loads to the upper end face, each delivering the same total axial force $F_{app} = 62.83$ kN:

1. **Uniform traction:** σ_{app} across the entire end face (the ES 120 idealization).
2. **Centered point load:** the full force at the single center vertex of the end face (think of a pin loading the bar from the center).
3. **Outer-ring traction:** a uniform traction applied only on the outer annulus ($r > 0.9 R_C$), representing grip-jaw loading that grabs the perimeter.

For each loading, I sampled the axial stress at nine cross-sections along the bar, clustered near the loaded end. On each cross-section, I used COMSOL’s Surface Average to compute the mean and

mean-square axial stress, $\langle \sigma_{yy} \rangle$ and $\langle \sigma_{yy}^2 \rangle$, and from those the standard deviation and the coefficient of variation:

$$\eta(y) = \frac{\sigma_{\text{std}}(y)}{|\langle \sigma_{yy} \rangle(y)|}, \quad \sigma_{\text{std}}(y) = \sqrt{\langle \sigma_{yy}^2 \rangle(y) - \langle \sigma_{yy} \rangle^2(y)}. \quad (1)$$

η is dimensionless: it equals zero for a perfectly uniform stress distribution (the ES 120 idealization) and grows as the distribution becomes more variable across the cross-section.

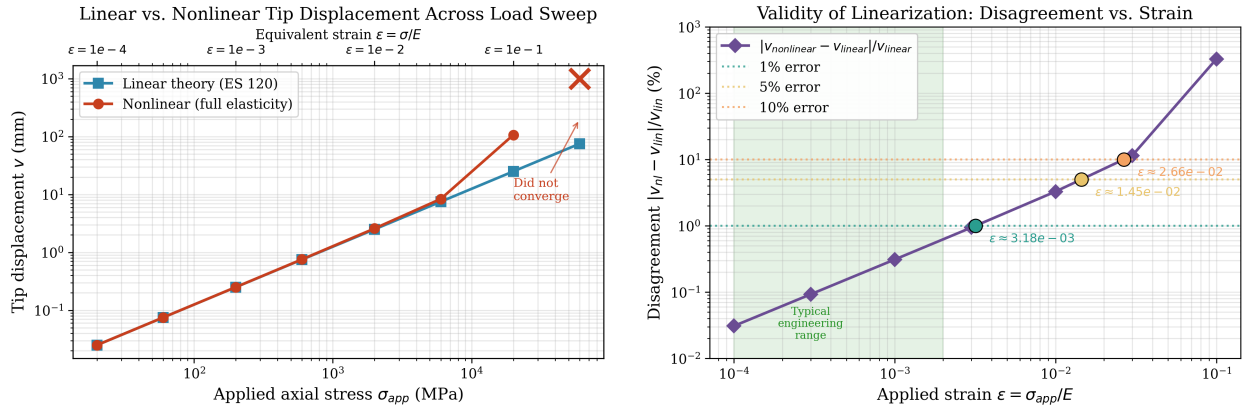
3 Data

3.1 Assumption I

Tip displacements under both formulations and their percent disagreement are shown in Table 1 and plotted in Figure 3.

Table 1: Linear vs. nonlinear tip displacement. Disagreement is $|v_{\text{nl}} - v_{\text{lin}}|/v_{\text{lin}}$.

σ_{app} (MPa)	$\varepsilon = \sigma/E$	v_{linear} (mm)	$v_{\text{nonlinear}}$ (mm)	Disagreement
20	10^{-4}	0.02505	0.02505	0.03%
60	3×10^{-4}	0.07514	0.07521	0.09%
200	10^{-3}	0.25047	0.25125	0.31%
600	3×10^{-3}	0.7514	0.7585	0.94%
2000	10^{-2}	2.505	2.587	3.3%
6000	3×10^{-2}	7.514	8.377	11.5%
20000	10^{-1}	25.05	107.0	327%
60000	3×10^{-1}	75.14	—	no convergence



(a) Linear and nonlinear curves overlap at low load and diverge at high load. (b) Disagreement reaches 1% at $\varepsilon \approx 3.2 \times 10^{-3}$, 5% at $\varepsilon \approx 1.4 \times 10^{-2}$, and 10% at $\varepsilon \approx 2.7 \times 10^{-2}$.

Figure 3: Linear vs. nonlinear tip displacement (left) and percent disagreement (right).

The two formulations agree to better than 1% for strains up to about 3×10^{-3} , and disagreement grows steadily from there. At $\varepsilon = 0.1$ the disagreement is over 300%, and at $\varepsilon = 0.3$ the nonlinear solver fails entirely, suggesting no stable equilibrium exists at that strain.

3.2 Assumption II

The three loadings produce visibly different stress fields at the loaded end (Figure 4). The uniform load gives the clean tensile field; the point load creates an extreme local stress at one node; the ring traction creates a bullseye of stress on the outer rim.

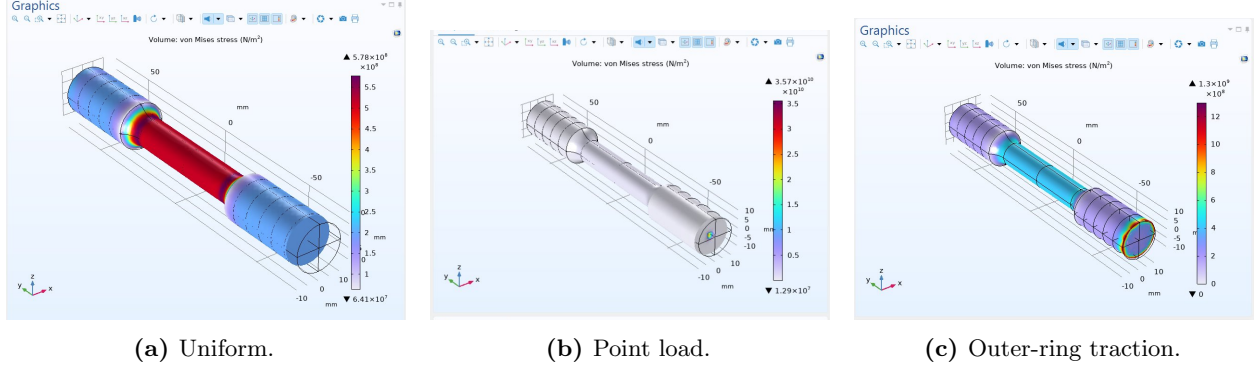


Figure 4: Von Mises stress for the three statically equivalent loadings.

The cross-sectional non-uniformity η is shown in Table 2 and plotted in Figure 5. Distance $d = y_{\text{load}} - y$ is measured from the loaded end face. Both the table and the plot show non-uniformity *outside* the fillet (rows inside the fillet are excluded for reasons explained below).

The point and ring loads start at $\eta \approx 21\%$ at the loaded end face and decay to a few percent within about one grip-diameter. The uniform load stays close to the noise floor ($\sim 4\%$) throughout. **This is Saint-Venant’s principle at work.** Statically equivalent loadings produce basically indistinguishable stress distributions sufficiently far from where the load is applied.

Fillet stress concentration. At the cross-section that falls inside the fillet ($y = 30$ mm and $y = 35$ mm, $d \approx 37$ mm–47 mm), I measured $\eta \approx 33\%$ for all three loadings. Because all three loadings give the same value, this is not a Saint-Venant effect — it’s an ordinary stress concentration at the change in cross-section between gauge and grip. I exclude these rows from Table 2 and Figure 5 to keep the focus on the Saint-Venant decay; the fillet effect is real but is a separate phenomenon requiring SCF analysis rather than Saint-Venant analysis.

Table 2: Coefficient of variation $\eta = \sigma_{\text{std}}/|\langle\sigma_{yy}\rangle|$ for the three loadings, outside the fillet. Position y is the cross-section’s y -coordinate; d is its distance from the loaded face. Entries marked “—” were not sampled for that loading.

y (mm)	d (mm)	η Uniform	η Point	η Ring
72.4	0.1	—	20.6%	20.9%
72.0	0.5	—	14.9%	15.4%
71.0	1.5	—	5.8%	6.7%
70.0	2.5	3.7%	4.2%	5.5%
65.0	7.5	—	4.1%	5.4%
55.0	17.5	—	1.7%	3.9%
45.0	27.5	2.8%	—	—
25.0	47.5	0.9%	2.7%	6.3%
0.0	72.5	$\sim 0\%$	2.1%	6.2%

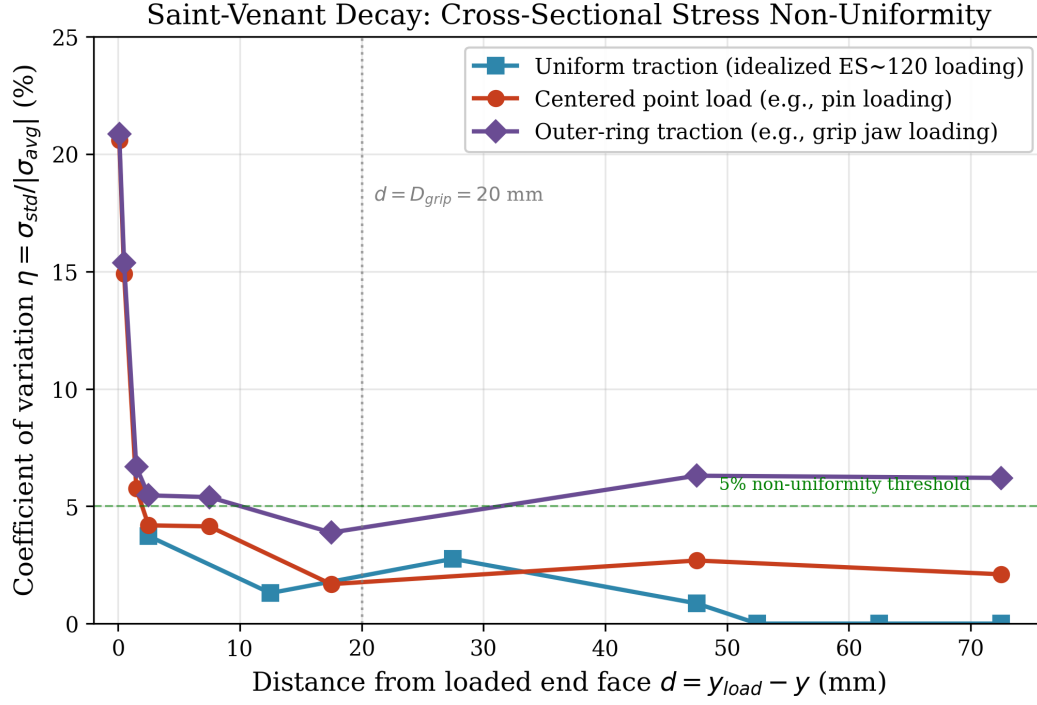


Figure 5: Cross-sectional non-uniformity η vs. distance from the loaded end (fillet excluded). The vertical dotted line marks one grip diameter, the canonical Saint-Venant length scale.

The mean stress on each cross-section is plotted in Figure 6. The uniform and point loadings hit the ES 120 predictions (512 MPa in the gauge, 200 MPa in the grip) within a fraction of a percent. The ring loading is offset down by about 9% due to a discretization effect explained in Section 5.

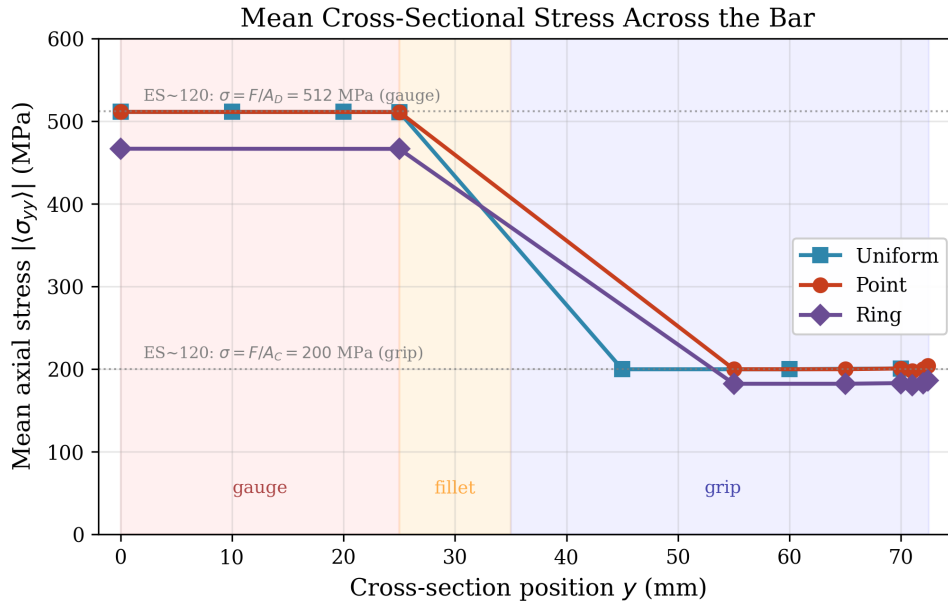


Figure 6: Mean axial stress along the bar. Uniform and point loadings match the ES 120 predictions (dotted lines) in the gauge and grip regions.

4 Assessment

Assumption I: Linearization

Linearization is valid for practical engineering use. The linear theory of ES 120 reproduces the full nonlinear result to within 1% for strains below $\varepsilon \approx 3 \times 10^{-3}$, and to within 5% for strains below $\varepsilon \approx 1.4 \times 10^{-2}$. Steel structures are typically designed at strains below 10^{-3} to keep a margin against yield, so linear theory has roughly an order of magnitude of strain margin in any realistic application. It only breaks down meaningfully beyond a few percent strain. This is past where steel itself has stopped behaving elastically.

Assumption II: Saint-Venant and Idealized Stress

It is valid in the gauge. Three very different boundary loadings produce stress distributions that look completely different at the loaded end face ($\eta = 4\%$ for uniform vs. 21% for point and ring), but they converge to within a few percent of each other within about one grip-diameter from the load. By the gauge center, the loadings are essentially indistinguishable. This means the use of $\sigma = P/A$ in the gauge of a tensile test is justified because the gauge sits outside the Saint-Venant zone of a realistic gripping arrangement.

Small caveat: ES 120 closed-form formulas should not be used across cross-sections that contain geometric features like fillets, holes, or notches. The fillet on this specimen creates $\eta \approx 33\%$ non-uniformity regardless of how the load is applied which isn't a Saint-Venant effect so that would likely require a different analysis.

Combined

Both assumptions are valid in the regime where ES 120 actually applies them. The full nonlinear, three-dimensional theory of elasticity adds nothing of practical importance to the linear analysis of a standard steel tensile specimen, except as a way to identify when the simpler theory breaks down: at fillets, near loads, or beyond the elastic limit.

5 Quick Notes

A couple of details worth noting:

1. The Uniform and Point runs in Assumption II were performed with geometric nonlinearity on (left over from Assumption I). This may have slightly affected the results, but not enough to render them invalid.
2. The peak stress under the point load (~ 36 GPa at one node) comes from applying a finite force at a single mesh node since cross-sectional averages integrate over the whole face.